

SQL Fundamentals Assignment

1. Retrieve all columns

SQL Query:

```
SELECT * FROM employees;
```

id	first_name	last_name	department	salary	hire_date	city
1	Alice	Green	IT	70000	2020-01-10	Johannesburg
2	Brian	Lee	HR	45000	2019-03-22	Cape Town
3	Cathy	Zulu	Finance	65000	2018-07-18	Durban
4	David	Mokoena	Marketing	50000	2021-11-05	Pretoria
5	Eva	Naidoo	IT	72000	2017-09-30	Johannesburg

2. Unique departments

SQL Query:

```
SELECT DISTINCT department FROM employees;
```

department
IT
HR
Finance
Marketing

3. Names ordered by salary DESC

SQL Query:

```
SELECT first_name, last_name FROM employees ORDER BY salary DESC;
```

first_name	last_name
Eva	Naidoo
Alice	Green
Cathy	Zulu
David	Mokoena
Brian	Lee

4. Top 3 highest-paid employees

SQL Query:

```
SELECT id, first_name, last_name, salary FROM employees ORDER BY salary  
DESC LIMIT 3;
```

id	first_name	last_name	salary
5	Eva	Naidoo	72000
1	Alice	Green	70000
3	Cathy	Zulu	65000

5. Employees in IT

SQL Query:

```
SELECT id, first_name, last_name, department FROM employees WHERE  
department = 'IT';
```

id	first_name	last_name	department
1	Alice	Green	IT
5	Eva	Naidoo	IT

6. Finance employees with salary > 60000

SQL Query:

```
SELECT id, first_name, last_name, department, salary FROM employees WHERE  
department = 'Finance' AND salary > 60000;
```

id	first_name	last_name	department	salary
3	Cathy	Zulu	Finance	65000

7. Employees in HR or Marketing

SQL Query:

```
SELECT id, first_name, last_name, department FROM employees WHERE  
department = 'HR' OR department = 'Marketing';
```

id	first_name	last_name	department
2	Brian	Lee	HR
4	David	Mokoena	Marketing

8. Employees NOT in IT

SQL Query:

```
SELECT id, first_name, last_name, department FROM employees WHERE  
department != 'IT';
```

id	first_name	last_name	department
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2	Brian	Lee	HR
3	Cathy	Zulu	Finance
4	David	Mokoena	Marketing

9. Employees in IT, HR, or Finance

SQL Query:

```
SELECT id, first_name, last_name, department FROM employees WHERE
department IN ('IT','HR','Finance');
```

id	first_name	last_name	department
1	Alice	Green	IT
2	Brian	Lee	HR
3	Cathy	Zulu	Finance
5	Eva	Naidoo	IT

10. IT employees salary > 65000 in Johannesburg

SQL Query:

```
SELECT id, first_name, last_name, department, salary, city FROM employees
WHERE department = 'IT' AND salary > 65000 AND city = 'Johannesburg';
```

id	first_name	last_name	department	salary	city
1	Alice	Green	IT	70000	Johannesburg
5	Eva	Naidoo	IT	72000	Johannesburg